

LAUNCESTON TI TREE BEND, Australia

WMO: 949690

Lat: 41.4194S

Lon: 147.1222E

Elev: 5

StdP: 101.26

Time Zone: 10.00 (AUS)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-0.8	0.2	-2.1	3.2	5.2	-1.2	3.4	5.0	11.0	12.7	9.6	11.7	0.6	100	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.2	28.2	18.8	26.4	18.2	24.8	17.3	20.2	25.5	19.3	24.3	18.5	23.1	5.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	13.4	22.0	17.6	12.6	21.4	16.6	11.8	20.8	57.9	25.5	55.1	24.3	52.4	23.2	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.2	7.3	DB	-2.3	32.1	0.9	2.2	-2.9	33.7	-3.4	35.0	-3.9	36.2	-4.6	37.8	(4)
(5)			WB	-2.3	22.1	1.2	1.0	-3.2	22.8	-3.9	23.4	-4.6	24.0	-5.5	24.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.9	18.4	18.4	16.6	13.2	10.4	8.3	7.9	8.7	10.2	12.2	14.7	16.4	(6)	
	DBStd	4.47	2.79	2.59	2.85	2.54	2.47	2.32	2.06	2.15	2.05	2.38	2.60	2.61	(7)	
	HDD10.0	243	0	0	0	4	26	62	70	53	22	6	1	0	(8)	
	HDD18.3	2073	33	28	71	156	245	302	323	299	243	189	113	71	(9)	
	CDD10.0	1309	261	236	204	99	39	9	6	12	28	75	141	199	(10)	
	CDD18.3	97	36	31	16	1	0	0	0	0	0	0	3	12	(11)	
	CDH23.3	714	282	191	101	2	0	0	0	0	0	3	37	98	(12)	
(13)	CDH26.7	135	67	31	17	0	0	0	0	0	0	0	3	17	(13)	
(14)	Wind	WSAvg	3.0	3.6	3.3	2.9	2.5	2.5	2.1	2.4	2.7	3.0	3.4	3.6	(14)	
(15) Precipitation	PrecAvg	669	52	29	34	54	67	68	78	82	65	55	49	45	(15)	
	PrecMax	784	139	107	106	125	139	183	158	162	116	121	85	145	(16)	
	PrecMin	493	10	0	3	17	8	21	37	15	8	17	6	6	(17)	
	PrecStd	84	32	24	29	28	35	33	31	44	28	27	19	30	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.7	29.9	29.0	23.2	18.8	16.2	15.0	16.5	19.4	23.2	26.9	29.0	(19)	
		MCWB	20.4	19.5	19.1	16.8	14.1	12.7	11.3	12.2	13.5	15.3	17.6	18.5	(20)	
	2%	DB	28.3	27.4	25.9	21.2	17.3	14.8	13.8	15.1	17.4	20.6	24.0	25.9	(21)	
		MCWB	18.8	19.0	18.0	15.6	13.3	11.9	10.7	11.4	12.3	14.2	16.9	17.5	(22)	
	5%	DB	26.3	25.6	24.0	19.7	16.2	13.8	13.0	13.9	16.1	19.0	22.0	23.9	(23)	
		MCWB	18.2	18.0	17.3	14.7	12.9	11.3	10.4	10.5	11.6	13.5	15.6	16.2	(24)	
	10%	DB	24.4	24.1	22.2	18.3	15.2	12.8	12.2	13.0	15.0	17.6	20.2	22.3	(25)	
		MCWB	17.1	17.2	16.2	14.0	12.3	10.8	10.0	10.1	11.1	12.6	14.5	15.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.9	21.1	20.3	18.0	15.4	13.8	12.4	13.3	14.6	16.7	19.1	20.1	(27)	
		MCDB	28.9	26.4	26.0	21.2	17.2	15.1	13.9	15.3	17.8	20.4	23.6	24.7	(28)	
	2%	WB	20.3	20.1	19.3	16.7	14.4	12.5	11.6	12.2	13.4	15.4	17.6	18.9	(29)	
		MCDB	25.9	25.2	24.1	19.6	16.2	14.0	12.9	14.0	16.2	19.0	22.4	23.7	(30)	
	5%	WB	19.2	19.2	18.2	15.9	13.7	11.8	11.0	11.4	12.5	14.3	16.5	17.8	(31)	
		MCDB	24.3	23.9	22.2	18.7	15.4	13.3	12.4	13.2	15.0	17.7	20.8	22.5	(32)	
	10%	WB	18.2	18.3	17.2	14.8	12.9	11.0	10.3	10.7	11.6	13.3	15.4	16.6	(33)	
		MCDB	23.1	22.7	21.1	17.5	14.8	12.6	11.8	12.5	14.1	16.8	19.4	21.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	10.9	10.8	10.4	9.5	9.1	8.8	9.0	9.6	10.3	10.4	10.7	(35)	
		MCDBR	13.6	12.7	12.6	11.9	10.0	9.4	9.1	9.8	10.9	12.0	12.9	13.1	(36)	
	5% WB	MCWBR	6.0	5.7	5.9	6.8	6.5	6.7	6.4	6.5	6.6	6.7	6.7	6.2	(37)	
		MCWBR	11.8	11.2	10.9	10.2	8.0	8.1	7.5	7.9	9.6	10.6	11.4	11.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.349	0.342	0.331	0.321	0.309	0.299	0.297	0.315	0.335	0.341	0.344	0.342	(40)	
		taud	2.480	2.520	2.545	2.548	2.559	2.575	2.586	2.532	2.463	2.455	2.471	2.489	(41)	
	Ebn at Noon		978	959	926	865	807	784	814	859	904	949	976	991	(42)	
		Edh at Noon	114	104	93	79	66	60	63	79	98	110	114	114	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.99	5.89	4.55	3.08	2.00	1.55	1.72	2.52	3.73	5.21	6.26	6.93	(44)	
	RadStd		0.47	0.24	0.20	0.16	0.12	0.14	0.08	0.20	0.26	0.27	0.45	0.45	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)	+0.32	N/A	N/A	+0.67	+0.38	+0.45	N/A	-100	+108	+14	

Nomenclature: See separate page